

MacNeille Reflection Checker Repair

Autonomous discussion Pass 111 verifies the Claude Code review repair for the finite MacNeille reflection checker.

For a finite preorder L , MacNeille completion is represented by lower cuts $C = (C^u)^l$. Since refutability $\boxtimes$ is antitone, the current checker extends it by first reading it as a monotone map $L \rightarrow L^{op}$. The current rule is

$$\widehat{\boxtimes}(C) = ((\boxtimes[C])^{l_L})^{u_L},$$

named **antitone-dual-lower-cut-v1**. The older **antitone-dual-lower-cut-v0** rule, $((\boxtimes[C])^{u_L})^{l_L}$, is retained only as a wrong-polarity comparison.

The decisive witness is **three-element-nolattice-nosynt**: carrier $\{0, a, b\}$ with $0 < a$, $0 < b$, and a incompatible with b ; refutability sends 0 to a and sends both a, b to 0 . It has no syntactic fixed point. Under v1, its MacNeille completion has the non-principal fixed cut $\{ 0, a, b \}$, classified as **nonprincipal-without-syntactic**. Under legacy v0, the same model gives $\{ 0, a \}$, principal at a but unreflected, and the principal extension condition fails twice.

The three-chain smoke test is also informative. Under v1 it has syntactic fixed point m , but the completed fixed cut is $\{ b, m, t \}$, principal at t and not reflected. Thus "principal" is not the same as "reflected"; the checker now records both **reflected** and **principal-unreflected** cases.

The audit report `artifacts/reports/pass111-macneille-reflection-review-check.json` verifies:

- the v1 non-lattice witness;
- the v0 wrong-polarity control;
- the v1 three-chain **principal-unreflected** smoke test;
- documentation markers for the repaired rule and classifications.

The result is PASS. The next mathematical task is to add APS axiom-package checks to the MacNeille reflection search and test whether any G2-holding finite model can keep a v1 non-principal completion fixed point without a syntactic fixed point.